



BAULKHAM HILLS HIGH SCHOOL

Assessment Task 2 2015
YEAR 11

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working
- Marks may be deducted for careless or badly arranged work

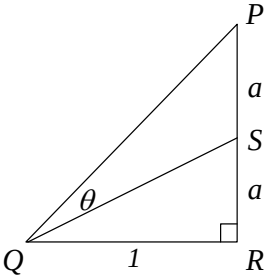
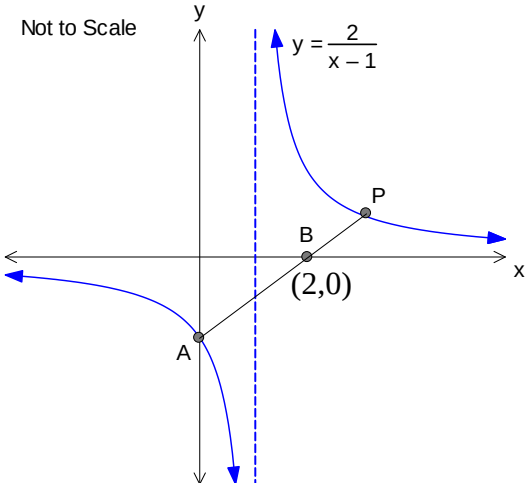
Total marks – 29

Exam consists of 3 pages.

This paper consists of TWO sections.

Question 1 (7 marks)		Marks
a)	Find the acute angle between the lines $2x + y = 4$ and $x - y = 2$, to the nearest minute.	2
b)	A is the point $(-4, 1)$ and B is the point $(2, 4)$. R is the point which divides AB externally in the ratio $2 : 1$. Find the co-ordinates of R	2
c)	If $\sin \alpha = \frac{3}{4}$, $0^\circ < \alpha < 90^\circ$ and $\sin \beta = \frac{2}{3}$, $90^\circ < \beta < 180^\circ$ find the exact values of $\sin(\alpha - \beta)$	3

Question 2 (8 marks)		
a)	Given $\cos \theta = x$. Express $\sin(270^\circ + \theta)$ in terms of x	2
b)	i) Express $\sqrt{3} \cos x - \sin x$ in the form $R \cos(x + \alpha)$, where α is in degrees.	2
	ii) Hence, or otherwise, solve the equation $\sqrt{3} \cos x - \sin x = 2$ for $0^\circ \leq x \leq 360^\circ$	2
c)	Prove $\tan(45^\circ + x) = \frac{\cos x + \sin x}{\cos x - \sin x}$	2

Question 3 (8 marks)	Marks
a)  In the diagram, angle PRQ is a right angle, and PS and SR are both of length a cm. QR is of length 1 cm. Express $\tan \theta$ in terms of a	2
b) Solve for $7 \sin x - 4 \cos x = 4$ for $0^\circ \leq x \leq 360^\circ$ by using the substitution $t = \tan \frac{x}{2}$	3
c) It is given that A is the y-intercept of the graph $y = \frac{2}{x-1}$. B is $(2, 0)$ as shown in the diagram below.  The line AB is extended to intersect the curve at point P. Find the ratio in which P divides the interval AB.	3

Question 4 (6 marks)	
a) i) Evaluate $\cos 105^\circ$ in exact form. ii) Hence or otherwise, find the exact value of $\cos 1005^\circ$	2 1
b) Given that $\cos 3x = 4 \cos^3 x - 3 \cos x$, find the exact value of $\sin 18^\circ$	3
End of the Exam	

Q. a) $2x + y = 4$ $x - y = 2$

$$m_1 = -2$$

$$m_2 = 1$$

$$\tan \theta = \left| \frac{1 - (-2)}{1 + (-2)(1)} \right|$$

Imp

$$\tan \theta = 3.$$

$$\theta = 71^\circ 34'$$

b) $(-4, 1)$ $(2, 4)$

$$-2:1$$

Imp = -ve for external ratio
• correct internal division
ie $R(0, 3)$.

$$R = \left(\frac{-2 \times 2 - 4 \times 1}{-2 + 1}, \frac{1 \times 1 - 2 \times 4}{-2 + 1} \right)$$

$$R(8, 7)$$

c) $\sin \alpha = \frac{3}{4}$

α is acute

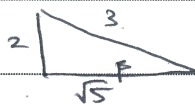


Imp • correct expansion
of $\sin(\alpha - \beta)$

$$\cos \alpha = \frac{\sqrt{7}}{4}$$

$$\sin \beta = \frac{2}{3}$$

β is obtuse



Imp • correct $\cos \alpha \cos \beta$

$$\cos \beta = \frac{-\sqrt{5}}{3}$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

$$= \frac{3}{4} \times \frac{-\sqrt{5}}{3} - \frac{\sqrt{7}}{4} \times \frac{2}{3}$$

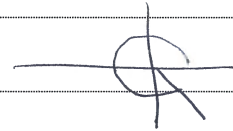
$$= \frac{-3\sqrt{5} - 2\sqrt{7}}{12}$$

Q2. a) $\cos \theta = x$

$$\sin(270^\circ + \theta) = -\sin(90^\circ - \theta)$$

$$= -\cos \theta$$

$$= -x$$



Imk acute $\angle = 90^\circ - \theta$.

Imk $-\sin \theta$.

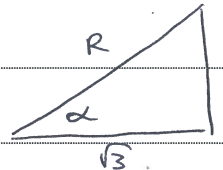
b)(i) $\sqrt{3} \cos x - \sin x = R \cos x \cos \alpha - R \sin x \sin \alpha$

$$R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = 30^\circ$$



$$R = 2$$

Imk. correct α or R .

$$\sqrt{3} \cos x - \sin x = 2 \cos(x + 30^\circ)$$

Imk correct expression with correct α and R .

(ii)

$$2 \cos(x + 30^\circ) = 2$$

$$+30^\circ < x + 30^\circ < 390^\circ$$

$$\cos(x + 30^\circ) = 1$$

Imk changing domain.



Imk exact value of $\cos \theta = 1$
 $\theta = 360^\circ$

* excluding $\theta = 0^\circ$.

$$x + 30^\circ = 360^\circ$$

$$x = 330^\circ$$

c) $\tan(45^\circ + x) = \frac{\tan 45 + \tan x}{1 - \tan 45 \tan x}$

Imk expanding $\tan x$.

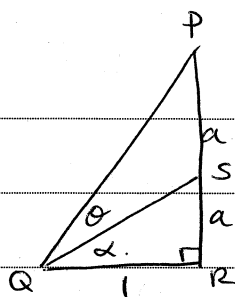
$$= \frac{1 + \frac{\sin x}{\cos x}}{1 - \frac{\sin x}{\cos x}} \times \cos x$$

$$= \frac{\cos x + \sin x}{\cos x - \sin x} \times \cos x$$

$$= \frac{\cos x + \sin x}{\cos x - \sin x}$$

$$= R.H.S.$$

Q3. a)



$$\tan \alpha = \frac{a}{b}$$

$$\tan(\theta + \alpha) = \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha}$$

Imp • expanding $\tan(\theta + \alpha)$
and using $\tan \alpha = 2a$

$$2a = \frac{\tan \theta + a}{1 - \tan \theta \times a}$$

$$2a - 2a \times a \tan \theta = \tan \theta + a$$

$$a = \tan \theta (1 + 2a^2)$$

$$\tan \theta = \frac{a}{1 + 2a^2}$$

Imp correct soln
for $\tan \theta$

b) $7 \sin x - 4 \cos x = 4$ for $0^\circ < x < 360^\circ$

$$\frac{7 \times 2t}{1+t^2} = \frac{4(1-t^2)}{1+t^2} = 4$$

Imp correct t formulae
in equation

$$4t^2 + 14t - 4 = 4 + 4t^2$$

$$7t = 4$$

$$\tan \frac{x}{2} = \frac{4}{7} \quad \text{for } 0 < \frac{x}{2} < 180^\circ$$

$$\frac{x}{2} = 29^\circ 45'$$

$$x = 59^\circ 29'$$

Imp = Finding x

When $x = 180^\circ$

$$7 \sin 180 - 4 \cos 180 = 4$$

$$\therefore x = 59^\circ 29', 180^\circ$$

Imp including
 $x = 180^\circ$

Q3 c) A(0, -2) B(2, 0)

∴ eqn of line AB is $y = x - 2$.

$$x - 2 = \frac{2}{x - 1}$$

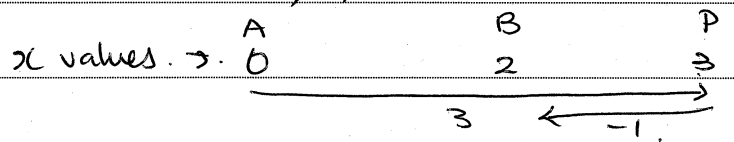
$$x^2 - 3x + 2 = 2$$

$$x(x - 3) = 0$$

$$x = 3$$

∴ P(3, 1)

Imp finding P.



∴ ratio is 3:-1 or -3:1.

Imp working towards finding Ratio.

Imp correct ratio with negative

or correct external ratio.

$$Q4. a) i) \cos 105 = \cos (60 + 45)$$

Imk expanding $\cos(60+45)$

$$= \cos 60 \cos 45 - \sin 60 \sin 45$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

Imk correct exact values

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{2}$$

$$ii) \cos 1005^\circ = \cos(105 + 180 \times 5)$$



$$= -\cos 105^\circ$$

$$= \frac{\sqrt{6} - \sqrt{2}}{2}$$

Imk correct soln.

$$b) \cos(3 \times 18) = 4\cos^3 18 - 3\cos 18$$

$$\cos 54 = 4\cos^3 18 - 3\cos 18$$

Imk correct eqn to find $\sin 18^\circ$

$$\sin 36 = 4\cos^3 18 - 3\cos 18$$

$$2\cos 18 \sin 18 = 4\cos^3 18 - 3\cos 18$$

$$0 = 4\cos^3 18 - 3\cos 18 - 2\cos 18 \sin 18$$

$$0 = \cos 18 (4\cos^2 18 - 2\sin 18 - 3)$$

$$0 = \cos 18 (4(1 - \sin^2 18) - 2\sin 18 - 3)$$

$$\therefore 4 - 4\sin^2 18 - 2\sin 18 - 3 = 0$$

$$4\sin^2 18 + 2\sin 18 - 1 = 0$$

Imk manipulating equation to something useful

$$\therefore \sin 18 = \frac{-2 \pm \sqrt{2^2 - 4 \times 4 \times -1}}{2 \times 4}$$

$$= \frac{-2 \pm \sqrt{20}}{8}$$

$$= \frac{-1 \pm \sqrt{5}}{4}$$

Since $\sin 18 > 0$

$$\therefore \sin 18 = \frac{-1 + \sqrt{5}}{4}$$

Imk correct value of $\sin 18^\circ$ and excludes negative value